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A system for FAIR assessment of computational biology models and simulations based on RDA indicators

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Computational models are essential for studying complex systems. We adapted the Findability, Accessibility, Interoperability, and Reusability (FAIR) indicators established by the Research Data Alliance to assess models encoded in domain-specific standards, including those developed by the COMBINE community. The resulting FAIR assessments guide FAIRification and, in turn, improve the transparent communication of a model's features, capabilities, and provenance, particularly in clinical settings.

Computational models are useful tools to study complex biological systems. Widespread translation of computational models into clinical practice is hindered by limited transparency of model code, lack of standardisation, insufficiently clear specification of model parameters, low reproducibility of simulation results[1] and a lack of model credibility[2]. In addition, data security and patient safety require models to be suitably quality-approved when used in a clinical setting. Publishing well-documented simulation studies alongside the journal publication has become part of good scientific practice within the COmputational Modelling in BIology NEtwork (COMBINE)[3], and is encouraged by many journals. Specific checklists have been developed to increase the rate of reproducible simulation studies[1]. One further step towards improving credibility and reproducibility of scientific results is the commitment to the scientific data management principles of Findability, Accessibility, Interoperability and Reusability (FAIR)[4].

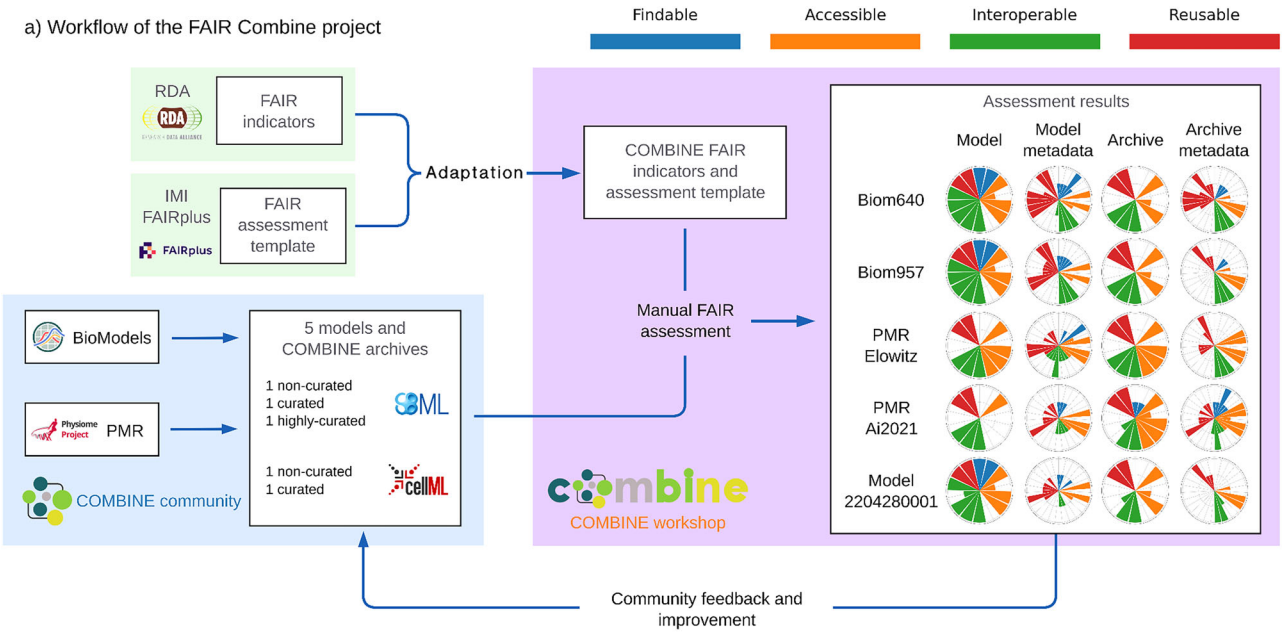
A FAIR assessment is a structured method to determine the extent to which a data object conforms to the FAIR guiding principles. A common set of domain-agnostic FAIR indicators has been published by the Research Data Alliance (RDA). Subsequently, the IMI FAIRplus project (https://fairplus-project.eu/, accessed April 2025) provided a template for FAIR assessment of data and metadata distinguishing three levels of agreement (0: no information/not fulfilled; 1: information fulfilled; and NA: not applicable)[5]. Depending on the application area, the subset of necessary RDA FAIR indicators should be identified. Notably, rich metadata is a key component in most assessments.

The fulfilment of all FAIR indicators requires substantial effort and expert knowledge. In most cases, a substantially higher FAIR score is easily reached if the already available information is structured and formatted correctly with support of FAIR experts. The results of a FAIR assessment, often visualised as so-called radial column charts, indicate the strengths and weaknesses of a dataset's FAIRness. An analysis of the radial columns leads to a FAIRification plan which then guides the steps towards improved FAIRness.

While an adequate adherence to the FAIR principles is the sole responsibility of model developers, repositories can define their minimal FAIRness requirements. Model repositories such as the BioModels[6] database (www.ebi.ac.uk/biomodels/, accessed November 2025) and the Physiome Model Repository[7] (PMR, https://models.physiomeproject.org, accessed November 2025) can be considered pioneers for transparency and reproducibility in scientific modelling.

To quantify the actual FAIRness of existing models, we launched the FAIR COMBINE Archive Indicators project (https://fair-ca-indicators.github.io/, accessed November 2025) as a community effort for the FAIR assessment of models and simulation studies within the COMBINE community. Building on our own previous assessments of biomedical data objects[8], we adapted the RDA FAIR indicators template provided by the

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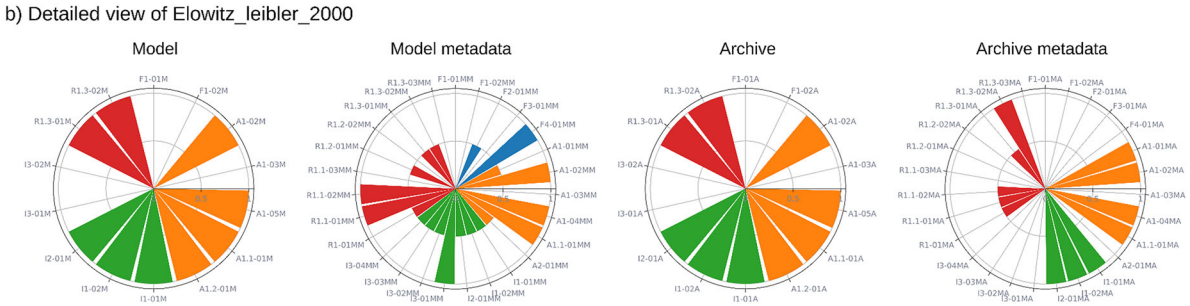


Fig. 1 | Workflow of the FAIR COMBINE Archive Indicators project. a An iterative and integrative community-level process to adapt the RDA FAIR indicators to the specificities of COMBINE models and archives. We extended the FAIRplus template of RDA FAIR indicators and performed FAIR assessment for 5 types of COMBINE models and archives from major model repositories (BioModels and PMR) at different curation stages. **b** FAIR assessment results for one model (https://models.physiomeproject.org/e/48/elowitz_leibler_2000.cellml). Results are shown for the resource itself (Model or Archive) and the metadata component (namely Model metadata or Archive metadata), for each FAIR indicator (Findability (blue), Accessibility (orange), Interoperability (green), Reusability (red)). The illustration is generated using the FAIR-Viz[27] tool (https://faircombine.streamlit.app/, github: https://github.com/matthiaskoenig/fair-ca-visualization). The complete FAIR assessment for this model is given in Supplementary Table 1.

FAIRplus project to cover the specificities of COMBINE simulations. Figure 1a summarises the workflow of the project, which was conducted under the umbrella of the RDA/European Open Science Cloud (EOSC) Future project (no. 101017536).

The COMBINE FAIR indicators were defined in a multi-step community process during two FAIR-dedicated workshops running at the main COMBINE conference and the corresponding hackathon in October 2022 and April 2023. These workshops were preceded by two other COMBINE workshops, held in October 2021 and April 2022, where we initiated the discussion about FAIR indicators. We organised online hackathons to revise the assessment (February–March 2023). Results were consolidated in project meetings by September 2024. We then consolidated 42 indicators for COMBINE models (which only capture the model formulation) and 42 indicators for archives[9] (which include supporting resources for the model). Details are provided in the Methods section.

As part of the EOSC-funded project, we initially evaluated the indicators on five different models and associated archives at different curation stages (Table 1). Specifically, we selected publicly available simulation studies which contained:

- The omex[10] standard format for the archive itself;
- Systems Biology Markup Language (SBML)[11] or CellML[12] files for model encoding;
- Systems Biology Graphical Notation (SBGN)[13] files for graphical network representation;
- Simulation Experiment Description Markup Language (SED-ML)[14] files for descriptions of simulation setups.

After publication of the initial assessment (January 2023), the established set of indicators has been further used and evaluated by the modelling community, e.g., the indicators were evaluated on five additional models by the PK/PD community including models on Dapagliflozin in Type 2 Diabetes Mellitus[15] or methods to physiologically based modelling of blood pressure regulation[16]. Assessment results have been made available together with a visualisation tool displaying the assessment results (https://faircombine.streamlit.app/[17]; assessments: https://github.com/FAIR-CA-indicators/FAIR-CA-indicators.github.io/).

Figure 1b summarises the assessment results for one of the models used during the workshops and hackathons (https://models.physiomeproject.org/e/48/elowitz_leibler_2000.cellml, accessed November 2025). We marked several exceptional cases with the score of 0.5 (i.e., between 0 and 1), where considerable work had already been done but the indicator was not yet fully satisfied. This partial score was assigned only when concrete

Table 1 | Overview of the five selected categories (for models and archives, considering the curation level): Non-curated model and curated model from BioModels; highly-curated BioModel, curated and non-curated model from PMR

No.	Model url	Model repository	Curation level	Comments
1.	https://www.ebi.ac.uk/biomodels/MODEL2204280001	BioModels	A non-curated model	Target: BioModels Archive metadata: xml file, metadata.rdf Model metadata: annotations within the model file, details from BioModels webpage
2.	https://www.ebi.ac.uk/biomodels/BIOMD0000000640	BioModels	A curated model	
3.	https://www.ebi.ac.uk/biomodels/BIOMD0000000957	BioModels	A highly-curated model from the BioModels COVID-19 collection that we built with previous EOSC Fast Track funding and that benefited from special curation following specific guidelines	
4.	https://models.physiomeproject.org/e/48/elowitz_leibler_2000.cellml/view	PMR	A non-curated model	Target: PMR repository Archive metadata: xml file Model metadata: PMR repository webpage, the elowitz_leibler_2000.session.xml and the annotations inside the model file.
5.	https://doi.org/10.36903/physiome.16828756	PMR	A curated model	Target: Physiome Journal Archive metadata: xml, report (in pdf format) accompanying the archive/ model in the journal Model metadata: the publication, the journal webpage, the report (pdf) and the SED-ML files. There are no annotations within the model files.

evidence of progress towards the indicator could be identified and the justification was documented in the assessment notes. The decision to apply the intermediate score was made following consensus among the FAIR and COMBINE experts participating in the workshops and hackathons.

Also, the score should be interpreted in the context of the same resource when evaluating improvements; the FAIR score is not informative for comparing models across resources. However, assessing multiple resources can reveal common gaps and trends across the community, providing useful insights for community-level improvements without implying direct numerical comparability.

The complete assessments and visualisation of all FAIRness scores are given in Supplementary Table 1 and Supplementary Fig. 1, respectively. A brief step-by-step walkthrough of the FAIR assessment process for the example model shown in Fig. 1b is provided in Supplementary Fig. 2.

Our assessments confirm that the COMBINE community has already achieved a reasonable FAIRness level. For example, the Reusability indicator on the model using community standards (CA-RDA-R1.3-01Model) scored highest (score = 1) because the models used the SBML and CellML standard formats. The model publication in BioModels and PMR also contributes to the FAIRness, because both provide API and protocols for accessing the models and archives, fulfilling the Accessibility indicators on automatic access and on free access protocol (CA-RDA-A1-05Model and CA-RDA-A1.1-01Model, score = 1). We see potential for further FAIR-ification of COMBINE metadata. Within the context of this work, metadata was partially available, but it was often distributed across several locations (including the XML file, the corresponding webpage or publication, within the models themselves) rather than being accessible in a single location, making it difficult to access information about the resources, especially in an automatic manner.

Based on the COMBINE FAIR indicators and lessons learnt (Table 1), we formulate the following recommendations for the provision of FAIR computational models:

1. **Metadata should be structured and standardised, should be provided in a separate file, and should remain available even if the computational resources are deleted**. Provide data and metadata in separate files (in agreement with the recommendation by Neal et al.[18]). The metadata needs to be both structured (to be machine-readable) and standardised (according to available community standards[18]). Separate provision of archive metadata will enable exploration and comparison of models and archives, without having to parse the full file set. In addition, the metadata should remain available even if the resources have been deleted.
2. **Quality of metadata should be assessed in addition to its pure existence**. Provide informative metadata with high information content[19]. Data objects with rich metadata may still contribute little to the actual understanding of the data if, for example, the metadata is not domain-specific, is not correct, or is too general.
3. **Basic provenance should be recorded**. Document changes to models, archives and metadata files using domain-specific provenance standards to make the longitudinal history of simulation setups available. We consider the metadata of simulation studies particularly essential for understanding model evolution, including information on the parametrisation, calibration and instructions for code execution[10].
4. **Each relevant component of a simulation study should have a unique ID**. Use IDs. As a consensus of this working group, we strongly recommend the consequent use of IDs (indicator CA-RDA-A1-01). Each model, model version, metadata file, and the archive itself should have their own IDs to ensure a consistent reference point, to track version changes, to refer to individual model versions, and to be able to put files into relation, even across archives.

The COMBINE community is committed to open and transparent science. In line with this mission, we encourage researchers to publish their FAIR baseline scores, so that the scientific community is not hampered by a feeling of unfairness, when they score low themselves. Reaching a high

FAIRness score (and everything above 80% is high) is a team effort, and requesting support from FAIR experts is highly recommended.

Throughout the project, we have disseminated our working strategies, results, lessons learned and challenges to the broader Open Science community.

In order to facilitate FAIR assessments, we developed a semi-automatic FAIR assessment tool for COMBINE models and archives, where some indicators are automatically assessed based on the properties of the inspected resources (models or archives). The prototype is openly available under Apache License 2.0 (backend[20]: https://github.com/FAIR-CA-indicators/fair-ca-indicators-backend, accessed November 2025; frontend[21]: https://github.com/FAIR-CA-indicators/fair-ca-indicators-frontend, accessed November 2025).

Finally, the presented list of COMBINE-adapted RDA FAIR indicators will guide the formulation of community guidelines for the FAIRification of models and archives. We believe that this effort is the first step towards a systematic FAIR assessment, leading to more FAIRness and thus faster curation and better reproducibility of computational modelling studies of biological systems. With this project, the interaction between RDA and COMBINE communities were strengthened: the COMBINE community benefits from reusing the existing RDA FAIR indicators and the RDA became more prominent within the COMBINE community.

Methods

In September 2022, we launched the FAIR COMBINE Archive Indicators project under the umbrella of the RDA/EOSC Future project to adapt the existing RDA FAIR indicators[6] to the field of computational biology models and archives[5]. Together with the COMBINE community, we adapted the RDA FAIR Data Maturity Model[5] indicators, following the template provided in the IMI FAIRplus project (https://fairplus-project.eu/, accessed November 2025), to the specifics of the COMBINE communities. All results have been published in the online project repository (https://fair-ca-indicators.github.io/, accessed November 2025).

FAIR-related work prior to the FAIR COMBINE archive project

Prior to the FAIR COMBINE Archive Indicators project, we performed a FAIR assessment using the RDA FAIR indicators in several biomedical projects[8], including the IMI BIOMAP (https://biomap-imi.eu/, accessed November 2025) and the EU ERAPerMed HeartMed (https://heartmed.pages.uni.lu/, accessed November 2025) projects. In addition, we were participants of the IMI FAIRplus project. We also coordinated the FAIR assessment[22] using the RDA indicators for the biological diagrams available in the Molecular Interaction NEtwoRk VisuAlization platform (https://minerva.pages.uni.lu, accessed November 2025) for the Parkinson's disease map (https://pdmap.uni.lu, accessed November 2025) and the COVID-19 Disease Map (https://covid.pages.uni.lu/, accessed November 2025), within the Disease Maps Project (https://disease-maps.io/, accessed November 2025).

Within the COMBINE community, we organized two FAIR-dedicated workshops in October 2021 and April 2022, where we conducted a series of introductory presentations and discussions about the FAIR principles, in general, and about the use of RDA FAIR indicators, in particular, with regard to the FAIR assessment of computational models. We also discussed with the community about benefits and issues of performing the FAIR assessment with respect to model quality. We then analysed the set of FAIR indicators provided by the RDA Data Maturity Model[5]. Generally, these indicators can be used in the context of COMBINE, but some degree of adaptation was needed to address the specifics of the COMBINE archives.

Community workshops for adaptation of RDA FAIR indicators to COMBINE

We organized a series of community workshops and meetings at which we asked for community contribution, dissemination, and consensus on (i) adapting the RDA FAIR indicators to the COMBINE models and archives, (ii) performing the FAIR assessment for several types of computational resources considering their curation level, and (iii) discussing further needs and steps. We coordinated the development of community guidelines for FAIRification of COMBINE simulations and developed a semi-automatic software tool to facilitate the FAIR assessment, given the input settings of the models and archives.

COMBINE 2022. At the annual community meet-up, COMBINE 2022[23], we organized a workshop specifically for the FAIR assessment of COMBINE archives. We aimed to raise awareness for FAIR in the COMBINE community and to specifically promote the assessment of the quality of research outcomes using a coordinated FAIR assessment. Circa 200 community members were present at the COMBINE 2022 meeting and were informed about the existence and possible use of FAIR assessment tools, specifically the RDA indicators. The workshop itself attracted 30 participants (developers and curators of COMBINE models, experts and users of the BioModels and PMR open model repositories) who discussed the first version of RDA indicators for FAIR assessment of COMBINE archives. Participants worked in five groups, each group focusing on the assessment of a pre-selected COMBINE model and archive:

1. A non-curated model from BioModels: https://www.ebi.ac.uk/biomodels/MODEL2204280001, accessed February 2025;
2. A curated model from BioModels: https://www.ebi.ac.uk/biomodels/BIOMD0000000640, accessed November 2025;
3. A highly-curated model from the BioModels COVID-19 collection that we built with previous EOSC Fast Track funding and that benefited from special curation following specific guidelines: https://www.ebi.ac.uk/biomodels/BIOMD0000000957, accessed November 2025;
4. A non-curated model from PMR: https://models.physiomeproject.org/e/48/elowitz_leibler_2000.cellml/view, accessed November 2025;
5. A curated model from PMR: https://doi.org/10.36903/physiome.16828756, accessed November 2025.

During this workshop, participants decided to perform the FAIR assessment on the COMBINE archive instead of assessing the COMBINE model itself (which captures the model formulation only). We continued working with the RDA FAIR Data Maturity Model[5] and evaluated the applicability of every indicator for the COMBINE archive. The workshop participants also explored the priority levels of the indicators.

Participants familiarized themselves with the RDA FAIR indicators and the FAIRplus template in smaller groups. Not all groups were able to finalize the assessment, as for most participants, this was the first time they had been exposed to the RDA FAIR indicators. The participants worked consciously and practically on the indicators. An important learning goal was the understanding that the model building and publication processes benefit from the FAIR indicators as early as during the model development stage. One key open question, which was addressed and decided upon in the later workshops, was whether to perform the FAIR assessment at the level of individual models (computational formulation file only), or archives (aggregated resources including the model itself and supporting files), or both in order to maximize the benefits.

Online hackathons. While the previous workshops were well-perceived by the community and received much attention in the COMBINE community at large, it became obvious that a detailed FAIR evaluation of the COMBINE archive demands much more work on the conceptual and implementation level. Importantly, there was a community-consensus that having the FAIR assessment for one category only (archives or models) would still miss specifics of the other category. Thus, we decided to systematically perform the FAIR assessment targeting both the COMBINE models and archives. We organized a bi-weekly series of five online hackathons between February and March 2023 to revise and finalize the assessment of the five example models (see above). Hackathons were open to the community and accompanied by an open call for feedback via GitHub issues (see details at https://github.com/FAIR-CA-

indicators/CA-RDA-Indicators/issues, accessed November 2025). Each session of the hackathon was a moderated one-by-one manual FAIR assessment of the chosen model and archive, respectively, based on the latest version of indicators. The working groups included participants with expertise on FAIR, COMBINE as well as on the BioModels and PMR model repositories.

The applied FAIRplus template includes 41 RDA FAIR indicators with 3 levels of priority (Essential, Important and Useful) and a binary scoring system: 1 meaning that a specific indicator was fulfilled, 0 otherwise. If the indicator is not specific to the given context, it is marked as not-applicable (NA). The FAIRplus template also distinguishes between Data and Metadata as key components.

We first mapped the FAIR components to the COMBINE resources as follows: the models and the archives (including all files within the archive) were considered as *Data* in the FAIR template, and the metadata of models and archives were considered as *Metadata*. Consequently, the Data-related indicators were renamed Model indicators and Archive indicators, respectively. Similarly, the indicators for Metadata were renamed Model Metadata indicators and Archive Metadata indicators, respectively. Thereby, the initial set of 41 RDA FAIR indicators had to be expanded to 82 indicators. 41 of these indicators focus on the Model and its metadata. The remaining 41 indicators focus on the Archive and its metadata. We anticipated the need for metadata to follow a machine-understandable cross-community standard in order to permit automatic retrieval and interrogation of the COMBINE resources (based on their metadata) from outside of the COMBINE community. Consequently, we proposed the introduction of 2 new Reusability indicators on the metadata of the archive and of the model, respectively, to comply with cross-community standards, namely: "CA-RDA-R1.3-03MA: Metadata of archive is expressed in compliance with a machine-understandable cross-community standard" and "CA-RDA-R1.3-03MM: Metadata of the model is expressed in compliance with a machine- understandable cross-community standard". Thus, in total, we defined a set of 84 FAIR COMBINE indicators. As part of the community consensus, we discussed and finalized the priority and the description of each indicator together.

The original FAIRplus template uses binary scores (0 or 1). However, we agreed to allow the intermediate score of 0.5 in exceptional cases to acknowledge existing information even if not completely satisfying the indicator. This scoring was agreed by consensus among the FAIR and COMBINE experts involved in the workshops and hackathons. For example, the requested information might be provided in a different form than the one expected by the indicator's description. In the assessment of the PMR curated model and archive, the Reusability indicator CA-RDA-R1.2-01MM ("Metadata of model includes provenance information according to community-specific standards") was scored value 0.5, with the explanation that some standardized metadata information had been available in the PMR Journal and in the repository itself. However, the metadata was not available in separate files. This approach targets the use of the intermediate score to documented transitional situations rather than ambiguous cases. A partial scoring strategy is expected to encourage the COMBINE community members to apply the FAIR principles. We anticipate that some of the current cases will more easily be solved once the community evolves. We also plan to discuss our partial scoring situation with other ongoing FAIR initiatives.

During our FAIR assessment sessions, we also noticed a level of dependency between indicators, where the assessment of the former ones as non-fulfilled would automatically determine the assessment of specific latter ones to non-fulfilled. For example, if an archive does not include any references to other data, it is evident that the archive will not include qualified references either. Thus, if the CA-RDA-I3-01Archive indicator on "Archive includes references to other data" is assessed to non-fulfilled (score = 0), then the CA-RDA-I3-02Archive indicator on "Archive includes qualified references to other data" can be automatically assessed as non-fulfilled (score = 0). Furthermore, we observed that some FAIR indicators are supported by the model repositories and can also be assessed automatically. For example, the BioModels repository has an overall license that must be accepted during model submissions. Therefore, the RDA indicators on the license (e.g., CA-RDA-R1.1-01MM) can be completed automatically for models stored in BioModels. We discussed the next steps, including the development of a semi-automated FAIR assessment tool to support the community in the FAIR assessment process, mainly by (1) automating the assessment for a subset of related indicators and (2) providing a visual interface to the assessment results and the suggested improvements.

HARMONY 2023. We continued work on the adaptation of the RDA FAIR indicators to the COMBINE simulations and we introduced a prototype of the semi-automatic evaluation tool. The development of the tool integrated the feedback from the community and continued until the end of the project.

Alignment with the European Open Science Cloud (EOSC) and the Research Data Alliance (RDA) initiatives

A key challenge for EOSC is to develop broadly applicable guidelines, tools, and workflows and to support their cost-effective specialization for community-specific needs[24]. The application of the broader RDA FAIR indicators to COMBINE standards is a prime example for this strategy. Our main challenge was achieving a community-level accepted set of FAIR indicators to capture the specifics of the COMBINE resources. For this, we relied on the community members being proactive. We continuously involved the community by organizing workshops, communicating transparently and acknowledging involvement, or via online dissemination of preliminary results. Moreover, we introduced our project goals as well as discussed, our strategies and results in several scientific conferences and meetings in the broader Computational Biology and Open Science community (including the ICSB 2022, EOSC Symposium 2022, FAIRDOMHub User Meeting 2023, 12th EURINT INTERNATIONAL CONFERENCE 2023, 1st Conference on Research Data Infrastructure 2023, ISMB/ECCB2023). We made connections to other relevant initiatives and projects such as FAIR Principles for Research Software (FAIR4RS Principles: https://rd-alliance.org/group/fair-research-software-fair4rs-wg/outcomes/fair-principles-research-software-fair4rs, accessed November 2025), NFDI4-DataScience (NFDI4DS: https://nfdi4datascience.de/, accessed November 2025), NFDI4Health (https://nfdi4health.de/, accessed November 2025), etc. All our work components were open and freely-accessible and connected via our project website (https://fair-ca-indicators.github.io/, accessed November 2025) to zenodo (see also the dissemination material[9,25,26]) and github for code files. Our aim is to foster the connection between the EOSC/RDA FAIR and COMBINE communities as well as to contribute to raising the potential application of FAIR indicators in other domains towards Open Science.

Data availability

FAIR indicators, including revisions, were deposited into a public GitHub repository and are available at the following URL: https://github.com/FAIR-CA-indicators/CA-RDA-Indicators/. A Docker image of the backend of the semi-automatic tool for FAIR assessment is available for free reuse (Apache-2.0 license) at the following URL: https://github.com/FAIR-CA-indicators/fair-ca-indicators-backend.

Code availability

The FAIR indicators and matrices to score the FAIRness are available from the following GitHub repository: https://fair-ca-indicators.github.io/.

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Author contributions

I.B. and D.W. conceptionalised and designed the initial work. I.B., D.W., D.P.N. and M.K. were responsible for initial data acquisition. I.B., D.P.N., J.A.H.W., H.H., M.K., R.S.M.S., M.S. and D.W. performed the initial data analysis and interpretation and refined the FAIR indicators. All authors (I.B., D.P.N., D.W., J.A.H.W., F.A., T.G., V.G., H.H., M.K., N.R., A.R., R.S., R.H.M., K.B.H., M.S., V.S., D.W.) contributed to the evaluation and second refinement of the indicators. I.B. and M.K. applied the indicators to additional models. All authors have drafted the manuscript and substantively revised it. T.G. and M.K. developed the evaluation tools.

Competing interests

The authors declare no competing interests.

Additional information

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